

# Predicting and Optimizing Electric Vehicle Charging Stations

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***Abstract - The project will aim at making predictions of charger piles at the publicly charged electric vehicles. This study aims at increasing the demand of forecasting, better allocation of resources as well as enable energy planning to be smart. The project methodology involves three stages, which include data preprocessing, feature engineering, regression modeling, and interactive user interface. The data contains such areas as operation, environment, and time, and is cleaned, transformed finally normalized before the models are trained. Several regression models including Linear Regression, Random Forest, Decision Tree, and XGBoost are train and test based on mean absolute error, root mean square error and coefficient of determination. Based on the results of the model training, XGBoost is identified as the most effective model by its nonlinear patterns flexibility and accuracy to predict. Interactive Streamlit interface is built to establish insight in comparison of models and makes real-time predictions. The results indicate that machine learning can be successfully used to model the charging behavior and give precise usage projections which enable the distribution of energy effectively and station management.***

***Index Terms - electric vehicle charging, machine learning, regression prediction models, usage forecasting***

## INTRODUCTION

The need to have quality and effective charging stations has increasingly become a significant challenge due to increase in the number of electric vehicles. Specifically, the problem of fluctuation in the pattern of usage is troublesome to the public charging stations taking into consideration such factors as locations, hourly usage, and

external conditions. The failure to make good predictions leads to congestion, poor allocation of energy, long queues, and wastage of operations in these stations.

Nevertheless, it is not an easy task because the usage of electric vehicle charging stations depends on numerous factors such as, but not confined to: the time of charging, electro-energy consumption, weather, time, and traffic observances. Therefore, as opposed to a conventional forecasting mechanism, machine learning offers a reasonable alternative to this issue, as it allows detecting the patterns automatically, and making high-accuracy forecasts using past data. That is, machine learning models will be able to give valuable answers about charging habits and predict the future demand of this type of charging stations [4].

This project is aimed at creation of a machine-learning based system which predicts the use of electric vehicle charging piles based on historical data. Throughout the analysis of the charging trends, and the determination of the determinant factors that affect the demand, the system seeks to assist the station operators in the maximization of energy distribution, avoiding traffic jams, offering a better experience to the drivers, and supporting the formation of sustainable transportation.

## RELATED WORKS

Previous studies have shown that charging infrastructure has continued to expand alongside the growth of electric vehicles. Arias and Bae [2] presented one of the early large-scale forecasting models in 2016, where they utilized big data technologies to analyze historical traffic and weather conditions to estimate electric vehicle charging requirements; their work demonstrated

that contextual variables influence charging behavior, and that integrating heterogeneous data sources can substantially enhance prediction quality.

Additionally, recent research has begun to incorporate deep learning techniques regarding the topic of the project as well. In the United Kingdom, Cavus et al. [1] developed a hybrid deep learning framework, HCB-Net, which combines neural networks with gradient boosting to forecast charging demand across different UK regions. As a result, their findings showed that hybrid architectures outperform standalone models for urban-scale prediction, and that factors such as regional mobility behavior, travel patterns, and spatial dynamics play a critical role in accurately forecasting charging demand. The two studies mentioned above were region-based or charging station-based studies, and van Etten et al. [3] proposed a global time-series modeling approach capable of predicting charging demand across large networks of stations simultaneously. Their work demonstrated improved generalization performance and highlighted the advantages of unified modeling in large-scale infrastructure planning.

After the analysis of related works, an emphasis was placed on the importance of taking contextual and temporal factors into consideration when predicting electric vehicle charging demand. Following the findings of these studies, this project implements a regression-based framework to predict charging pile usage using historical operating and environmental features.

## **Methodology**

The section on the methodology explains the procedure employed in making the prediction of the electric vehicle (EV) charging demand based on historical data obtained on the electric vehicle charging stations. The methodology is further broken down into some important components data collection, data cleaning and preprocessing, feature engineering, machine learning model creation, model evaluation metrics and lastly the creation of an interactive Streamlit user interface to help view results and offer real time predictions.

### ***A. Data Collection***

The data in terms of this study has been taken as a publicly available dataset called Charging Pile Demand Forecasting

Dataset on Kaggle, which is comprised of operational data detailed in electric vehicle charging stations. It records various attributes to the use of charging piles, which include timestamps, charge time, amount of energy used, assigned vehicles charging, traffic flow, busiest time of the day, type of vehicle, temperature, and price incentives. Specifically, this dataset may be applied in predicting the charging demand since it unites some temporal data (timestamps), environmental (e.g., temperature) and operational (e.g., amount of consumed energy and vehicles number) data. With the help of these characteristics, machine learning models will be able to comprehend the multifaceted relationships between these variables and give precisely correct demand predictions.

### ***B. Data Cleaning & Preprocessing***

Data cleaning and preprocessing are the necessary steps to make sure that the dataset is appropriate to calculate and be used in machine learning. Preprocessing is carried out as follows: Missing values are frequent occurrences, and it mainly happens in large datasets. Where continuous variables are used, e.g. energy consumption or charging period, the missing values are statistically imputed by means or median imputation. In the case of categorical variables, the missing data is substituted by the most repeated or a new category is created labeled as Unknown.

In the case of the dataset used in this study, no missing values were found. So, there was no need for substituting any data. However, there were irrelevant columns in predicting the demand of charging and are not shown in the dataset as they are unique identifiers of the stations or irrelevant metadata. To simplify the dataset so that it does not cause noise in the models, these columns are dropped.

As for time series analysis, time-tamps were crucial in determining peak demand at certain times of the day. Based on this, the time and day of the week are obtained to record the daily and hourly trends in the charging behavior. The reason for dropping the month and year from the dataset, was that the data collected were all recorded in the same month and year. As an example, demand could peak at certain times of the day, e.g. in the evening and during the busiest commute periods. This action aids this model in capturing the temporal features in a better way.

Machine learning models often demand features to be of a similar scale that is going to be useful. Such attributes as energy consumption, traffic flow, and charging duration should be standardized i.e. scaled to zero mean and unit variance resulting that all characteristics should make equal contribution to the prediction task. It is especially crucial with models such as Linear Regression and XGBoost in which different scales of features may significantly impact performance over representatively.

### ***C. Feature Engineering***

The creation of new features or changes to the existing features with a view to improving the predictive capability of the model is referred to as the process of feature engineering. The engineering techniques of feature estimation include Duration per Electric Vehicle, Energy per Minute, and Rolling Usage Rate.

Duration per EV will be a new feature that will record the average time spent charging an electric vehicle. It is determined by dividing the total number of hours taken by a fair number of vehicles charging in that time. This characteristic enables the model to know the relationship between charging time and the total station demand, which may also be applicable in predicting the peak hours of demand [5]. Energy per Minute is the rate of energy consumption per minute during charging. It is obtained through the division of the total energy used and the total charging time. This aspect also gives the idea of the efficiency of charging stations, and this can be significant towards forecasting the upcoming energy needs.

Rolling usage rate is calculated as a moving average of the usage of the station within a certain time interval. This characteristic has been effective in smoothening the demand fluctuations and detecting the long-run tendencies. It is especially applicable in predicting demand that will increase or decrease depending on historical trend.

### ***D. Machine Learning Models***

In this research, several machine learning models are trained and tested to determine the charging demand at electric vehicle stations.

One of the simplest regression models is Linear Regression. It supposes the linear dependency between the

input features and the target variable. Though it is computationally inefficient, it might not render complicated, non-linear associations in the data, particularly in case of such factors like weather, traffic and time. Random Forest is a type of ensemble learning which perceives numerous decision trees, thus capable of prediction. It is stronger than single-decision trees and has the capability of modeling non-linear associations on the data [7]. Random Forest is also applicable when one has a large dataset of many features since it helps eliminate overfitting by averaging the forecast of numerous trees.

Decision Tree is a non-linear model, and the model divides the data into subsets based on the values in the features. It is simple to interpret, and the tree structure is easy to understand however it can be easily over-fitted to noise in the data. Thus, the pruning algorithm or such an ensemble approach as Random Forest can be the favorite choice. Finally, XGBoost model is a non-linear model and the comparison of the values of the features alone forms the subsets of the data. It is easily interpretable, and the tree structure is easily comprehensible, but it may be readily over-fitted with noise in the data. So, such an ensemble algorithm as Random Forest or pruning can be the preferred option [8].

### ***E. Model Evaluation Metrics***

The results of the machine learning regression models are assessed in the following metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and  $R^2$  (Coefficient of Determination).

Mean Absolute Error is one of the measures used to assess the average performance when comparing the predicted and actual values. It is an easy-to-use measure of model accuracy, which directly measures prediction errors [9]. Root Mean Squared Error (RMSE) is also very similar to MAE, but it stresses more on big errors. It also punishes models with large errors more, which is why it is a convenient metric to use to evaluate models in situations where large errors pose a special problem.

$R^2$  (Coefficient of Determination) is a measure that is used to estimate the amount of variability in the target variable that can be explained by the model. The larger the value of  $R^2$  the more the model explains the data.  $R^2$  comes in

incredibly handy when it comes to comparison of the performance of various regression models.

### F. Streamlit User Interface

Streamlit is a Python library intending to develop data applications using minimal code. An interactive Streamlit dashboard is created so that other users can interact with the model and investigate the predictions that it makes.

This dashboard includes real-time predictions, users can enter the existing data (e.g. temperature, traffic, time of the day) and obtain real-time forecasts on how much the station would charge their car. Additionally, it proposes the comparison of various machine learning models (Linear Regression, Random Forest, Decision Tree, XGBoost) by their evaluation values (MAE, RMSE, R2), giving the user the ability to select the most appropriate one.

The dashboard also includes a range of different charts to represent the data regarding the trends in charging demand, peak time of use, and other valuable information such as bar graphs, line graphs, and heat maps. These graphics aid the stakeholders in making decisions which are informed about the management of the stations and the energy allocation.

### Summary of Main Insights

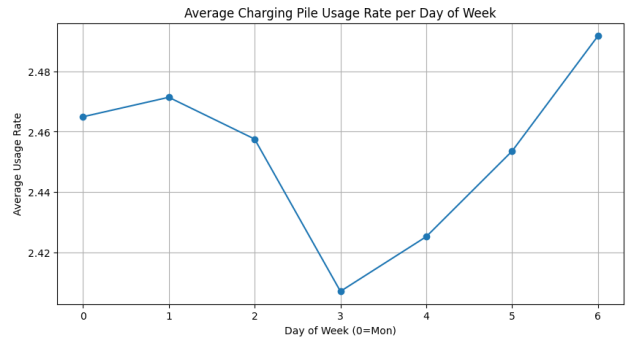
The most important predictors for charging demand were time of the day and traffic flow; these are the factors which have significant impact on the timing and quantity of the energy used in the stations.

The best performing model was XGBoost; the other models failed to perform with regards to prediction. This is also because it has the capacity to represent complex, non-linear associations within the information and because it can handle extensive quantities of characteristics. XGBoost worked well compared to Linear Regression and Decision Tree, which failed to build such relationships.

Although the results from Linear Regression were the best and almost perfect, it does not actually show how well the model is performing, as the dataset has simple relationships and features explain the target extremely well. Thus, the researchers have decided to remove Linear

Regression out of the equation when accounting for best model performance.

It was visualized that the demand peaks in peculiar hours (5AM-6PM) of the day meaning when there is peak charging, and there was an increase in pile usage rate during the weekend.



The analysis findings are relevant in the practice of EV stations operators. Knowledge of peak charging periods and external conditions will enable operators to distribute the energy optimally and minimize congestion at the charging points. Also, the real-time prediction service will be useful in enabling the operators to manage their stations better so that they are equipped during high demand time.

### Conclusion

Use of machine learning in prediction of a charging of an electric vehicle has been effective. It was demonstrated that the models could represent the demand of charging on the basis of the historical data with the precise approval, which enables allocating the energy and managing the stations more productively. This is not the only way such a strategy can enable the optimization of the existing charging infrastructure but will also contribute to the creation of progressive and efficient EV infrastructure.

Machine learning and regression models may be crucial to simplify the operation of charging stations by including rational predictive signs on the trends of demand. This can accomplish better allocation of resources, constrained congestion and a more sustainable utilization of energy. Also, the higher accuracy of the predictions due to the provision of real-time data streaming and bigger datasets may facilitate an even more precise energy distribution strategy and the demand forecasting.

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